Lecture 1: Introduction to Financial Machine Learning

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Overview of Financial Machine Learning

- **What is Financial Machine Learning?**
  - Tailored to financial applications.
    - Non-Stationary Data, Low Signal-to-Noise Ratio, Extreme Events, Tail Risks, Regularization
  - Combines data-driven insights with economic theory.
    - ML as a complementary tool to existing economic models
    - ML as a "discovery tool," revealing insights that challenge or expand existing economic theories, driving theoretical innovation.
    - Incorporate Economic Theory into ML Algorithm Design
    - ML algorithms become interpretable, robust, and aligned with economic reasoning
- **Why Financial Machine Learning?**
  - Traditional models are limited in handling complex market dynamics.
  - ML allows for adaptive, data-rich decision-making.

Prices Are Predictions

Information Set

Functional Forms

Machine Learning vs Econometrics

Challenges of Applying ML in Finance

Two Cultures of Financial Economics

Definition of Asset Prices

$$P_{i,t} = \mathrm{E}\left[M_{t+1}X_{i,t+1} \mid \mathcal{I}_t\right]. \tag{1.1}$$

- Asset price ($P_{i,t}$) reflects:
  - Future payoffs ($X_{i,t+1}$).
  - Discounting based on marginal substitution rates (M_{t+1}).
- Price reflects investors' valuation of its future payoffs. These valuations are discounted based on investors' preferences.
- **Interpretation**: *Prices are predictions - they reflect investors' predictions for the (discounted) future payoffs shed by an asset based on available information.*
- Example: Suppose a company is expected to pay dividends of $D_{t+1} = 2$ \$ per share next year, and investors' expected marginal rate of substitution (discount factor) is $M_{t+1} = 0.9$. Then $P_{i,t} = 1.8$ \$. If new information (e.g., a positive earnings report) leads investors to predict higher dividends ($D_{t+1} = 3$ \$), then $P_{i,t} = 2.7$ \$.

Expected Return Representation

$$\mathrm{E}\left[R_{i,t+1} \mid \mathcal{I}_t\right] = \beta_{i,t}\lambda_t. \tag{1.2}$$

- Components:
  - $R_{i,t+1} = \frac{X_{i,t+1}}{P_{i,t}} - R_{f,t}$: Excess return. including price changes and dividends.
  - $R_{f,t} = \mathrm{E}[M_{t+1} \mid \mathcal{I}_t]^{-1}$: Risk-free rate.
  - $\beta_{i,t}$: Asset's covariance with M_{t+1}.
  - λ_t: Price of risk.
- **Interpretation**: Expected Returns reflect investors' expectations for the appreciation in asset value over the next period.
- Higher β: The asset is more sensitive to market fluctuations and requires a higher expected return.
- Higher λ: In riskier market environments, investors demand higher compensation, raising expected returns.

Connection

$$P_{i,t} = \mathbb{E}\big[M_{t+1}\, X_{i,t+1} \mid \mathcal{I}_t\big] \longrightarrow 1 = \mathbb{E}\big[M_{t+1} R_{i,t+1}^{(\mathrm{gross})} \mid \mathcal{I}_t\big],$$

where $R_{i,t+1}^{(\mathrm{gross})} = \frac{X_{i,t+1}}{P_{i,t}}$. Next, define the one-period risk-free rate as

$$R_{f,t} = \mathbb{E}[M_{t+1} \mid \mathcal{I}_t]^{-1}.$$

So $\mathbb{E}\big[M_{t+1}(R_{i,t+1}^{(\mathrm{gross})} - R_{f,t}) \mid \mathcal{I}_t\big] = 0$, i.e. $\mathbb{E}\big[M_{t+1} R_{i,t+1} \mid \mathcal{I}_t\big] = 0$, where $R_{i,t+1} = R_{i,t+1}^{(\mathrm{gross})} - R_{f,t}$ is expected excess return. Then

$$\mathbb{E}[R_{i,t+1} \mid \mathcal{I}_t] = -\frac{\mathrm{Cov}\big[M_{t+1}, R_{i,t+1} \mid \mathcal{I}_t\big]}{\mathbb{E}[M_{t+1} \mid \mathcal{I}_t]}.$$

We define

$$\beta_{i,t} = \frac{\mathrm{Cov}[M_{t+1}, R_{i,t+1} \mid \mathcal{I}_t]}{\mathrm{Var}[M_{t+1} \mid \mathcal{I}_t]} \quad \text{and} \quad \lambda_t = -\frac{\mathrm{Var}[M_{t+1} \mid \mathcal{I}_t]}{\mathbb{E}[M_{t+1} \mid \mathcal{I}_t]},$$

thus obtaining

$$\mathbb{E}[R_{i,t+1} \mid \mathcal{I}_t] = \beta_{i,t}\, \lambda_t.$$

Returns vs Prices

Stock	P_t	X_{t+1}	**Expected Return** (R_{t+1})
A	\$50	\$55	$\frac{55}{50} - 1 = 10\%$
B	\$200	\$220	$\frac{220}{200} - 1 = 10\%$

- **Advantages of Returns**:
  - Stationary properties ease statistical modeling.
  - Normalize scale differences across assets.
- **Why Prefer Returns?**
  - Homogenize data for tractability.
  - Critical for resource allocation and investment decisions.
- Both prices and returns provide predictive insights.
- **Dual Role of Models**:
  - Enhance scientific understanding of markets.
  - Improve real-world decision-making (e.g., portfolio management).

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Importance of Information in Finance

- **Core Question**: "What information do market participants have, and how do they use it?"
- Prices reflect information about:
  - Future asset payoffs ($X_{i,t+1}$)
  - Investor preferences (M_{t+1})
- The study of asset prices is tied to understanding how information shapes market behavior.

Large Conditioning Information Sets

- Vast scope of conditioning information:
  - Structured Data: historical stock prices, trading volumes, etc. Macroeconomic indicators such as GDP growth, interest rates, and inflation.
  - Unstructured Data: news articles, social media, earnings calls, expert predictions, etc.
- Investors have access to varying types and levels of information.
  - Retail Investor: Publicly available data
  - Institutional Investor: Premium and proprietary data sources
  - High-frequency trading data, Alternative datasets, Paid financial models

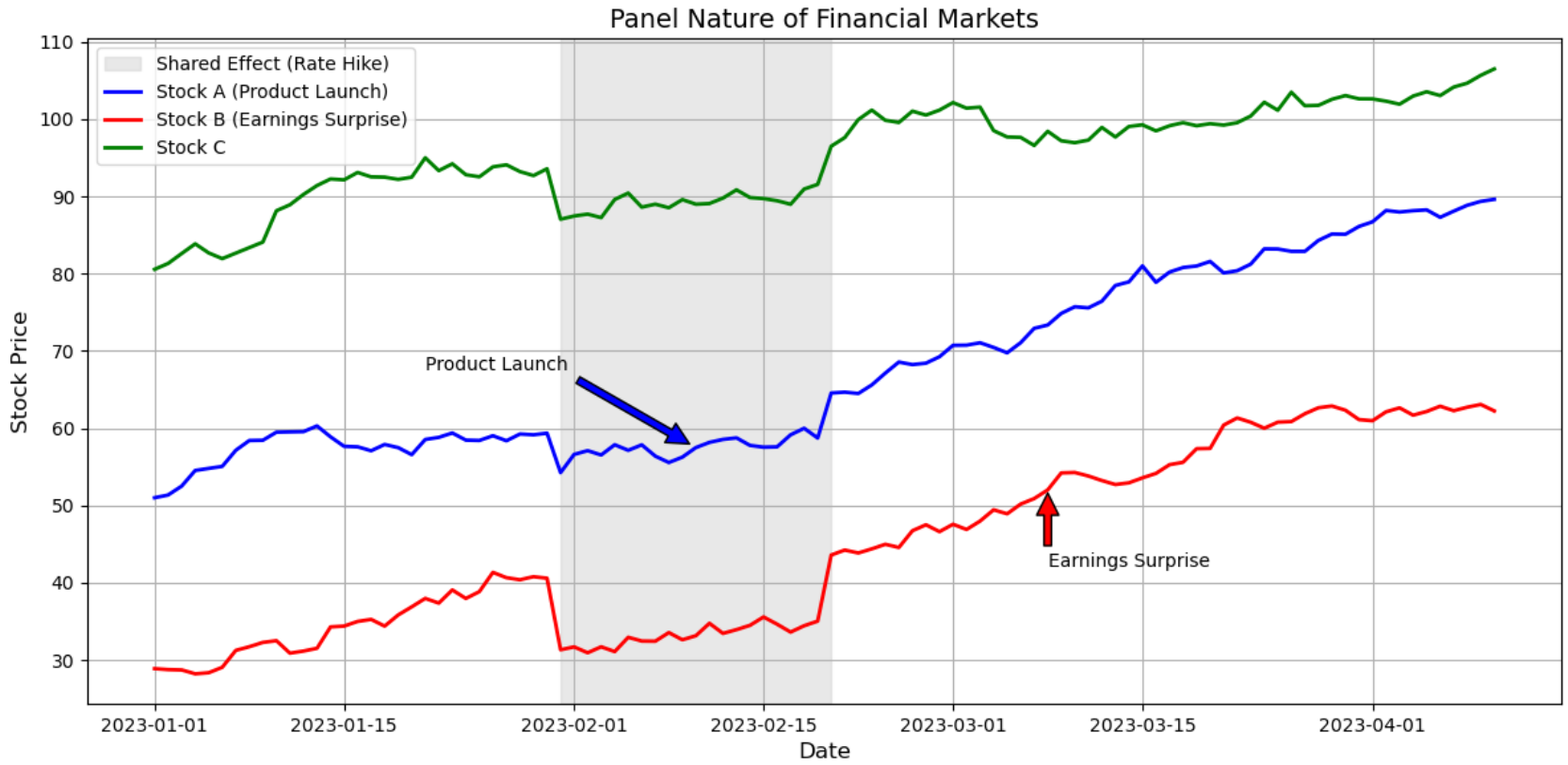
Panel Nature of Financial Markets

- **Time Series Dimension**: Variation in asset prices over time.
  - The price of Stock A increases steadily due to improving company fundamentals and earning reports.
- **Cross-Section Dimension**: Differences in prices across assets at a given time.
  - On the same day, Stock A increases 5%, Stock B increases 1%, and Stock C decreases 3%, reflecting their distinct conditions and potentials.

A successful model must simultaneously account for

1. **Shared Dynamic Effects**:
   - Market-wide influences (e.g., risk conditions, policy changes).
2. **Asset-Specific Effects**:
   - Static or dynamic attributes unique to individual assets or asset groups.

Example of Panel Nature



Simultaneous modeling of:

- Shared market effects.
- Asset-specific attributes.

Challenges in Modeling Asset Prices

Financial economics research: investigate one or a few variables at a time

- **Predictors in Literature** (Gu et al. [2020]):
  - Hundreds of stock-level predictive characteristics.
    - Size, Value, Momentum, etc
  - Dozens of macroeconomic predictors.
    - market-wide trends
    - Interest Rates, GDP Growth, Employment Rates
- Emerging Frontiers:
  - Expanding unexplored information sources.
  - Incorporation of news text into empirical models.
  - Challenges: Data Quality, Dynamic Relationships, Dimensionality

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Role of Functional Forms in Finance

- **If asset prices are expectations of future outcomes, then the statistical tools to study prices are forecasting models.**
- Traditional Structural Models:
  - Specify a functional form based on economic theory.
  - Estimate parameters to link information to market prices.
  - Example: Capital Asset Pricing Model (CAPM)
  - The CAPM specifies that the expected return of a stock is a linear function of its sensitivity to market risk (β):

$$\mathrm{E}\left[R_{i,t}\right] = R_{f,t} + \beta_{i,t}\left(\mathrm{E}\left[R_{m,t}\right] - R_{f,t}\right)$$

- **Challenges**:
  - fails to match market data.
  - often lacks flexibility and are evaluated primarily in-sample.

Reduced-Form Models and Machine Learning

- Shift away from structural assumptions to:
  - **Reduced-Form Models**: Emphasize empirical patterns without economic constraints.
  - Example: Fama-French Models
  - Assumptions: Linearity, Low-dimensional factor models.
  - Challenge: Many choices for statistical structure and assumptions.
  - subjective choice: "right" statistical structure (e.g., which factors to include)
- Opportunity: Machine learning offers flexibility and scalability
  - Nonlinearity, High Dimensionality, Time-varying
  - Integrate economic theory with flexible modeling techniques

Flexible Approaches to Modeling

- **Machine Learning Methods**:
  - Kernel methods (e.g. SVM)
  - Penalized likelihood estimators (e.g. LASSO, RIDGE)
  - Decision trees
  - Neural networks
- Designed to approximate unknown data-generating functions.
- Can integrate diverse data sources ("Multimodality") into a single model.

Information and Functional Forms

- **Rich Conditioning Information**:
  - Cochrane [2009]: Observed information is only a fraction of what market participants use.
- **Information Gaps**:
  - Econometricians face limitations compared to market participants.
- Machine learning narrows the gap by assimilating large information sets.
- Same logic for functional form
- Investors use information in potentially complex ways that are difficult to be captured by parametric models.

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Definition and Scope of Machine Learning

- **Definition** (Gu et al. [2020]):
  - High-dimensional models for statistical prediction.
  - Regularization methods for model selection and avoiding overfit.
  - Efficient algorithms for searching across diverse model specifications.
- **Key Insight**:
  - Machine learning and econometrics share statistical foundations.
  - Machine learning emphasizes prediction accuracy over hypothesis testing.

Functional Representation in Machine Learning

- **Representation Choice**:
  - Small models: Parsimonious and interpretable, but rigid.
  - Large models: Flexible and capable of capturing complexities, but prone to overfit.
- **Nonparametric Approach**:
  - Machine learning pursues statistical analysis without strict assumptions.
  - Open-minded exploration of model specifications guided by data.
- **Key Feature**:
  - Maximizing prediction accuracy in the face of unknown data-generating processes.

Model Selection and Regularization

- **Model Selection**:
  - Core to machine learning.
  - Searches through a diverse collection of candidate models to find top performers.
- **Regularization**:
  - Encourages stable performance out-of-sample.
  - Balances model complexity to avoid overfitting.
  - Large enough to capture complex relationships.
  - Small enough to avoid overfitting noise.

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Small Data in Economic Time Series

- **Data Scarcity**:
  - Macro finance data limited to a few hundred observations.
  - Time series data accrues slowly with time.
- **Comparison to Other Domains**:
  - Machine learning in other fields benefits from virtually unlimited data.
  - Financial ML must adapt to limited data environments.

Weak Signal-to-Noise Ratios

- **Market Efficiency**:
  - Profit maximization and competition reduce return predictability.
- **Source of Price Variation**:
  - Predominantly driven by unanticipated news.
    - unforecastable noise from the perspective of the model
  - Time-varying risk premia and inefficiencies provide limited predictability.
    - we should expect return predictability to be small and fiercely competed over.
- **Impact on ML Models**:
  - High noise complicates learning and prediction.

Structural Instability in Financial Markets

- **Evolving Markets**:
  - Predictive patterns may be arbitraged away.
    - Pre-2000s: Many investors relied on simple momentum strategies
    - Post-2000s: the rise of algorithmic and high-frequency trading (HFT)
  - Regulatory and technological changes reshape markets.
- **Moving Target**:
  - Models must adapt to dynamic environments.
    - in times of market crashes or regime shifts (e.g., 2008 Financial Crisis), historical patterns and relationships between predictors no longer hold.
- **Compounded Challenges**:
  - Small data and high noise are exacerbated by instability.

Benefits of Economic Structure

- **Role of Theory**:
  - Theory reduces the parameter estimation burden.
  - Provides a filter for noise, enhancing signal extraction.
- **Hybrid Approach**:
  - Combine economic theory for structure.
  - Use machine learning to explore aspects beyond theoretical scope.
- **Efficient Data Use**:
  - Balance between model complexity and data limitations.

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Two Cultures in Financial Economics

- **Structural Model/Hypothesis Test Culture**:
  - Focuses on structural assumptions and hypothesis testing.
  - Employs constrained prediction models with limited functional forms and conditioning variables.
  - Generalizes poorly beyond the model's purview or training data.
- **Prediction Model Culture**:
  - Values statistical explanatory power above adherence to economic theory.
  - Employs flexible, reduced-form models, often using machine learning.
  - Produces robust data fits and identifies new empirical patterns.

Role of Prediction Models in Economics

- **Criticism and Misunderstanding**:
  - Often criticized for lacking direct economic theory linkage.
- **Scientific Advancement**:
  - Complement hypothesis testing by exploring broad empirical landscapes.
  - Facilitate theory development through new empirical insights.
- **Practical Benefits**:
  - Aid investors, consumers, and policymakers with improved predictions.
  - Enhance welfare by refining resource allocation decisions.

Statistical Challenges

- Real-world allocations are constrained by mis-specified and noisy models.
- All models are wrong, but some are useful.
- **Statistical Wedges**:
  - First-best: Perfectly specified and known models.
  - Second-best: Mis-specified models without noise.
  - Third-best: Mis-specified and noisy models, which are the reality.
- **Machine Learning's Role**:
  - Mitigates the gaps between theoretical optimality and real-world practice.
  - Provides improved predictions from complex, high-dimensional data.

Example 1: Predicting Stock Returns

Richard Roll (1987): By observing and measuring these influences ex post, one could presumably "explain" stock price movements with an astronomical degree of accuracy. The R^2 should be close to 1.0.

- **First-best**
  - If the true return-generating process were known, $R_{i,t} = f(\text{economic conditions, firm characteristics, investor behavior})$.
- **Second-best**
  - The researcher estimates a simplified linear model: $R_{i,t} = \beta_0 + \beta_1 \text{Size}_t + \beta_2 \text{Value}_t + \beta_3 \text{Momentum}_t$.
  - Limitation: The model misses nonlinearities and interactions.
  - If there were no noise, it could still provide useful explanations.
- **Third-best**
  - Returns are affected by "unobservable" factors (e.g., investor sentiment, global shocks).
  - Market dynamics change over time, making past models unstable.
  - High-dimensional interactions exist that traditional models fail to capture.

Example 2: Portfolio Choice

- **First-best**
  - Known expected returns and covariance matrix.
- **Second-best**
  - Estimate expected returns and covariances using historical data
  - assumes stable relationships
  - regime shifts, tail risks, nonlinear dependencies.
- **Third-best**
  - Investors estimate expected returns and covariance under uncertainty.
  - wrong parameter estimates + market noise
- **Machine Learning Contributions**:
  - Enhances return and risk forecasts for practical decision-making.
  - Provides scalable methods to manage complex datasets for portfolio optimization.
  - Robust portfolio optimization under uncertainty

Summary

- **Prices are Predictions**:
  - Asset prices reflect discounted expectations of future payoffs.
  - Expected return representation normalizes prices for better analysis.
- **Large Information Sets**:
  - Financial markets integrate vast, complex information.
  - Panel nature of financial data requires simultaneous modeling of time series and cross-sectional effects.
- **Ambiguous Functional Forms**:
  - Structural models face challenges of limited flexibility and generalizability.
  - Machine learning offers flexible approaches to address functional ambiguity.
- **ML vs Econometrics**:
  - Machine learning prioritizes prediction accuracy and model flexibility.
- **Challenges and Opportunities**:
  - Small data, weak signal-to-noise ratios, and evolving markets challenge ML applications.

Q&A

- Questions or thoughts about today's lecture?

John H Cochrane. *Asset pricing: Revised edition*. Princeton university press, 2009.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies*, 33(5): 2223–2273, 2020.